HOW THE RESEARCH WAS CONDUCTED

AI Collusion Research Sprint, 21 to 26 October 2026 (sprint: 23 to 25 October). Question: does a cartel among language-model pricing agents survive when a firm is replaced? All times IST. Dates and counts come from the repository's daily notes, results note and git log.

Phases: planning · frozen plan · build and pilot · main sweeps · exploratory additions and audit · final runs and report

Wed 21 Oct (before the sprint)

planning

- **Question chosen:** "cartel turnover", whether collusion survives when a firm is replaced (sprint question 1.2), with a fallback study. Repository, license and tag `pre-sprint` created; a local model server set up. No experiment code, runs or results.

Thu 22 Oct (before the sprint)

frozen plan

- **Reading:** seven related papers read (abstracts and introductions, then design sections); harness specified on paper; an exploratory ablation list written.
- **Plan frozen:** game, gate, swap branches and outcome fixed in the repository before any data (tag `prereg-v1`).

Fri 23 Oct (sprint day 1)

build and pilot

- **Sprint start** (tag `sprint-start`). Harness and tests written the same afternoon (game, agents, play, swaps, run, analysis).
- **Pilot gate: 5** local models x **3** seeds. Only gemma4-12b formed a cartel (3 of 3); qwen3-14b, gpt-oss-20b, mistral-nemo-12b and phi4-14b priced near the competitive level. Main swap study on gemma started that evening.

Sat 24 Oct (sprint day 2)

main sweeps

- **gemma4-12b swap study finished overnight:** seeds 11 to 20, four branches each, 9 of 10 runs formed a cartel; then two firms replaced and announced swaps.
- **Hosted endpoint added** (a change from the local-only plan): deepseek-v4-1-flash seeds 11 to 20, then qw38 and glm53f gate and swap studies. Messages off and history off: without messages cartels formed in 1 of 5 and 0 of 10 runs.
- **First exploratory cells:** communication cut at the swap, private notes, known end date, mixed-model and eight-firm markets.

Sun 25 Oct (sprint day 3)

exploratory additions and audit

- **Overnight and morning:** gap-fill cells, awareness probes (do firms notice a swap?), instructed newcomer prompts, 120-round horizon, 2, 4 and 5 firms, entry of new firms.
- **Hosted credit ran out** during the glm53f newcomer cells (21 of 25 branches finished). No hosted runs after that. Round-1 spread test on **50** seeds.
- **Human case studies: 5** games with the author as one firm, then a **4**-game information suite (one game per setting, not estimates).
- **Evening audit** by four read-only reviews (leaks, numbers, documents, report readiness), fixes applied, and the git history rewritten once to remove material not meant to be public (logged in the disclosure log). Report draft rebuilt as V2.

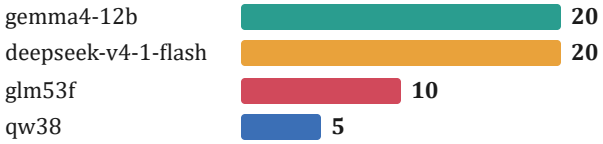
Sun 25 to Mon 26 Oct

final runs and report

- **Overnight, local GPU only:** gemma seeds 31 to 40 added (now 20 seeds, 17 of 20 formed a cartel; no direction changed), swap timing (swap at round 15, 30 or 45), and qwen3-14b as a second local base model, which never formed a cartel (0 of 7 runs) and so cannot test persistence.
- **Report written** on the official template; tag `submitted` on the submitted commit.

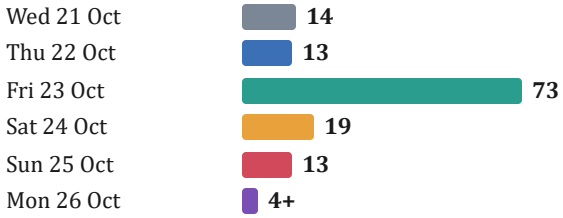
Effort, as recorded

Base runs in the main swap study, by model



Each base run is 30 rounds followed by several 30-round branches from its checkpoint. Not shown: pilot (15 runs), factor, mixed, horizon and newcomer cells, round-1 test (50 seeds), human games (9).

Commits per day (git log)



136 commits up to midday 26 Oct; Monday is still in progress. Hosted spend: about 36 USD of a 50 USD credit by the end of the mixed and eight-firm runs (results note, section 16).

Honesty notes

Deviations are logged. Every change after the frozen plan (models, the hosted endpoint, token limits, every ablation, the human games, extra seeds, swap timing, the second base model) is dated in the deviations section of the frozen plan and treated as exploratory.

Not run (cost or time): a 20-round history window, reasoning-effort and anchor-price ablations, swap studies on a gate-only hosted model, factor cells for qw38, swap timing on the hosted models, and further hosted runs once the credit ran out.

How the work was done. One author. Code and analysis were written with an AI coding tool; local runs used a single consumer GPU, hosted runs a hosted endpoint.